

| PhD thesis in Signal Processing

Title *Bayesian Estimate of Galaxy Clusters for Improved Cosmological Parameter Inference*
Location *Centre de Recherche en Informatique, Signal et Automatique de Lille (CRISTAL, SigMA team), Villeneuve-d'Ascq, France*
Supervisors [Jenny Sorce](#) and [Pierre-Antoine Thouvenin](#)
Funding ANR Chaire WILL UNIVERSITWINS Project hold by [Jenny Sorce](#) and [Klaus Dolag](#)
Keywords Inverse problem solving, Bayesian hierarchical model, generative models, MCMC algorithms.

| Research internships

Apr. 2025 - **Pre-PhD Research internship** *Centre de Recherche en Informatique, Signal et Automatique de Lille (CRISTAL, SigMA team), Villeneuve-d'Ascq, France*
Sep. 2025

Title *Bayesian Estimate of Galaxy Clusters for Improved Cosmological Parameter Inference*
Supervisors [Jenny Sorce](#) and [Pierre-Antoine Thouvenin](#)
Funding ANR Chaire IA SHERLOCK hold by [Pierre Chainais](#)
Keywords Inverse problem solving, Bayesian inference, X-ray spectra modeling.

Apr. 2024 - **Research internship**, *Centre de Recherche en Informatique, Signal et Automatique de Lille (CRISTAL, SigMA team), Villeneuve-d'Ascq, France*
Jul. 2024

Title *Galaxy cluster mass estimate using radial velocities*
Supervisors [Jenny Sorce](#) and [Pierre-Antoine Thouvenin](#)
Funding ANR Chaire IA SHERLOCK hold by [Pierre Chainais](#)
Keywords Inverse problem solving, physic based inference, Block Coordinate Forward Backward algorithm.

| Research projects

Title *Mapping the distance between the simulated and the observed cosmic webs*
Year Master 2
Supervisors [Jenny Sorce](#) and [Jérémie Boulanger](#)
Keywords Constrained simulations, separation metrics to define dissimilarities: likelihood, Mahalanobis and Sinkhorn distance.

Title *Galaxy cluster mass estimate using radial velocities*
Year Master 1
Supervisors [Jenny Sorce](#) and [Pierre-Antoine Thouvenin](#)
Keywords Inverse problem solving, pattern recognition, Gaussian Processes.

Title *Introduction to Topological Data Analysis (TDA)*
Year 3rd y. Bachelor
Supervisors [Camille Sabbah](#)
Keywords Basic notions of topology, persistent homology, mapper algorithm, Betti number.

🎓 | Education

2023 - 2025 **Master Degree in Data Science** (rank: 2/19), *Université de Lille, Lille, France*
Research-oriented Master entirely taught in english.
2020 - 2023 **Bachelor Degree in Applied Mathematics** (rank: 2/28), *Université de Lille, Lille, France*
Multidisciplinary bachelor focused on mathematics, economics and computer science.

| Experience in teaching

- Apr. 2026 - **Co-supervision of two Master students** *Centre de Recherche en Informatique, Signal et*
Sep. 2026 *Automatique de Lille (CRIStAL, SigMA team)*, Villeneuve-d'Ascq, France
- Year 1st y. PhD Thesis
- Topic Galaxy quenching, Virgo cluster, simulation-based inference
- Keywords Assistance on supercomputers, cosmological simulations, RAMSES, RDRAMSES.
- Oct. 2022 - **Mathematics Tutor of 1st y. Bachelor** *Mathématiques et informatique appliquées aux sciences*
Apr. 2023 *humaines et sociales (MIASHS)*, Université de Lille, Lille, France
- Year 3rd y. Bachelor
- Lectures Mathematical analysis and algebra
- Keywords Limits, continuity, differentiation, differential equation, vector spaces, linear maps.
- Apr. 2022 - **Intern in secondary education** *Collège Notre Dame de la Paix* and *Lycée Notre Dame de la*
Jul. 2022 *Paix*, Lille, France
- Year 2nd y. Bachelor
- Lectures Small lectures in mathematics (from Year 7 to Year 12), a larger one at Year 13 (2×6 hours).
- Keywords Arithmetics, geometry, statistics, probability, function analysis, law of large numbers.

| Languages and programming Skills

- Languages French (Native), English (C1 level), German (B1 level)
- Programming Python, Git, R, Git, LaTeX, SQL, HTML
- Softwares: PowerBi, Excel*